

Priyesh Vashistha

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Research Interests: Haptics and Tactile Sensing — Rehabilitation Robotics — Human-Robot Interaction — Modular Mechatronics — Robotics and Automation.

EDUCATION

Queen's University

MEng. Electrical and Computer Engineering

Kingston, ON

Sept 2025 – Present

- Coursework: Human-Robot Interaction, AI & Interactive Systems, Search & Planning Algorithms for Robotics, Machine Learning/Deep Learning, Robotics Engineering Research & Design, Engineering Innovation.

Bikaner Technical University

B.Tech in Electronics & Communication Engineering

Rajasthan, India

Sept 2020 – July 2024

- Coursework: Digital System Design, Microprocessors & Controllers, Computer Architecture, Digital Electronics, Electronic Devices & Circuits, VLSI Design.

JOB EXPERIENCE & INTERNSHIPS

Graduate Engineer Trainee (R&D Department)

Pyrotech Electronics Pvt. Ltd. (Unit-II)

March 2025 – Aug 2025

Udaipur, India

- Designed MCC and PCC panels using AutoCAD and SolidWorks; generated SLDs and electrical schemes.
- Standardized calculations and design rules for busbars and busducts across multiple electrical ratings.

Embedded System Intern

Sofcon India Pvt. Ltd.

May 2023 – Aug 2023

Ahmedabad, India

- Utilized Embedded C for hardware integration with AT89C51, PIC16F877A, and Arduino microcontrollers.

Machine Learning Intern

MaxGEN Technologies PVT. LTD.

July 2022 – July 2022

Pune, India

- Developed ML models using Logistic Regression, Random Forest, and XGBoost with SMOTE, achieving 86% accuracy and 0.86 ROC-AUC in customer churn prediction algorithms.

ACADEMIC RESEARCH & ENGINEERING PROJECTS

Tactile Sensing Smart Glove for Stroke Rehabilitation | Queen's University

May 2026 – Aug 2026

- Investigated kinematic sensing for stroke rehabilitation by engineering a custom multi-modal sensor glove to continuously track spatial orientation and evaluate fine motor movements.
- Engineered custom rigid and flexible printed circuit boards (PCBs) in Altium, integrating FSR 400 series pressure sensors, flex sensors, and multiple IMUs.
- Authored a comprehensive technical report (ELEC 898) detailing the sensor data acquisition methodology, 3D spatial modeling, and palmar clamshell mechanical design.

Dual-Drive Modular Microbot: Autonomous Latching | [GitHub](#) | Queen's University

Jan 2026 – Apr 2026

- Evaluated multi-agent latching mechanisms by prototyping a tethered dual-drive microbot to assess edge-triggered magnetic field detection as part of advanced graduate coursework (MECH 853).
- Validated kinetic mobility and sensing via an Arduino UNO validation platform using the Web Serial API for real-time teleoperation and A3144 Hall Effect sensors.

Multi-Heuristic A* Ablation Study: Mars Rover Navigation | [GitHub](#) | Queen's University

Sep 2025 – Dec 2025

- Conducted an ablation study evaluating heuristic diversity in MHA* algorithms for terrain-aware navigation on grid maps with active energy constraints.
- Demonstrated a quantifiable reduction in node expansions and planning time across complex simulated terrains.

Interactive Cardiovascular Risk Assessment System | [GitHub](#) | Queen's University

Sep 2025 – Dec 2025

- Designed a hybrid machine learning system (ROC-AUC: 0.802) featuring a deterministic "Safety Layer" to enforce strict medical rules and overrides.
- Conducted fairness audits on model outputs, demonstrating a negligible Equalized Odds Gap (0.010) across gender demographics.

Multi-Robot Coordination: Blame and Trust Attribution | [GitHub](#) | *Queen's University* Sep 2025 – Dec 2025

- Investigated how social role priority impacts human attribution of blame and trust during robotic system failures.
- Utilized animated 3D stimuli and psychological frameworks to analyze user perception of "competence cliffs."

Low Cost Electronic Ventilator | *Jodhpur Inst. of Eng. & Tech.* 2024

- Engineered an Arduino-based ventilator prototype with real-time monitoring, achieving an approximate 90% cost reduction compared to standard medical models.

Bluetooth/Voice Controlled + Object Sensing Robot | *Jodhpur Inst. of Eng. & Tech.* 2023

- Developed a 10cm x 10cm x 17cm robot using Arduino Uno, L293D motor driver, and ultrasonic sensors.
- Integrated Bluetooth module and voice control for navigation and automated object sensing.

SKILLS

Programming Languages: Python, C/C++, Embedded C

Hardware Platforms: Arduino, AT89C51, PIC16F877A, Microcontrollers & Processors

Embedded Systems: PCB Design, Circuit Design, Haptics & Kinematic Sensing

EDA & CAD Tools: Altium Designer, AutoCAD, SolidWorks, Autodesk Fusion, Bambu Studio

LEADERSHIP

President – ECE Dept. (2023-24) | Treasurer – ECE Dept. (2022-23) | Head – Robotics Club (2022-24)